CLOUD INFRASTRUCTURE & VIRTUALIZATION

Hypervisors, Virtual Machines, Containers, and Cloud Data Centre Architecture

1

Virtualization Concepts

Hypervisors, Type 1 vs Type 2, types of virtualization, benefits and limitations.

2

VMs vs Containers

Architecture, trade-offs, Docker, and how the two are combined in practice.

3

Cloud Data Centre Architecture

Regions, availability zones, resource pooling, multi-tenancy, orchestration.

BEFORE WE BEGIN

1

Explain virtualization and the role of the hypervisor

Describe how a hypervisor abstracts physical hardware to run multiple operating systems on one machine.

2

Differentiate Type 1 and Type 2 hypervisors

Identify their architecture, examples, performance trade-offs, and typical use cases.

3

Compare virtual machines and containers

Explain the architectural difference and choose the right technology for a given workload.

4

Describe cloud data centre architecture

Explain regions, availability zones, resource pooling, multi-tenancy, and orchestration.

INTRODUCTION

What is Cloud Infrastructure?

1

Definition

The hardware and software components — servers, storage, networking, and virtualization — required to support cloud computing.

2

Delivered over the internet

Resources are provisioned and accessed remotely, on demand, without the customer owning the physical equipment.

3

The foundation for cloud services

This infrastructure underpins IaaS, PaaS, and SaaS offerings from providers such as AWS, Azure, and Google Cloud.

PART 1

1

Virtualization Concepts

Hypervisors, virtualization types, and the trade-offs they introduce.

What is Virtualization?

1

Definition

Creating a virtual (software-based) version of a resource — a server, storage device, network, or operating system.

2

One machine, many machines

Virtualization lets a single physical machine behave as if it were several independent machines.

3

The core enabling technology of the cloud

Nearly every cloud service is built on top of a virtualization layer that shares physical hardware across many customers.

Life Before Virtualization

1

One application per physical server

The common practice through the 1990s and early 2000s: each server ran a single OS and a single application.

2

Very low utilization

Physical servers commonly ran at only 10–15% average CPU utilization, wasting the rest of the capacity.

3

Expensive to scale

Adding capacity meant buying, racking, powering, and cooling another entire physical machine.

What is a Hypervisor?

1

Also called a Virtual Machine Monitor (VMM)

Software that creates, runs, and manages virtual machines.

2

Sits between hardware and VMs

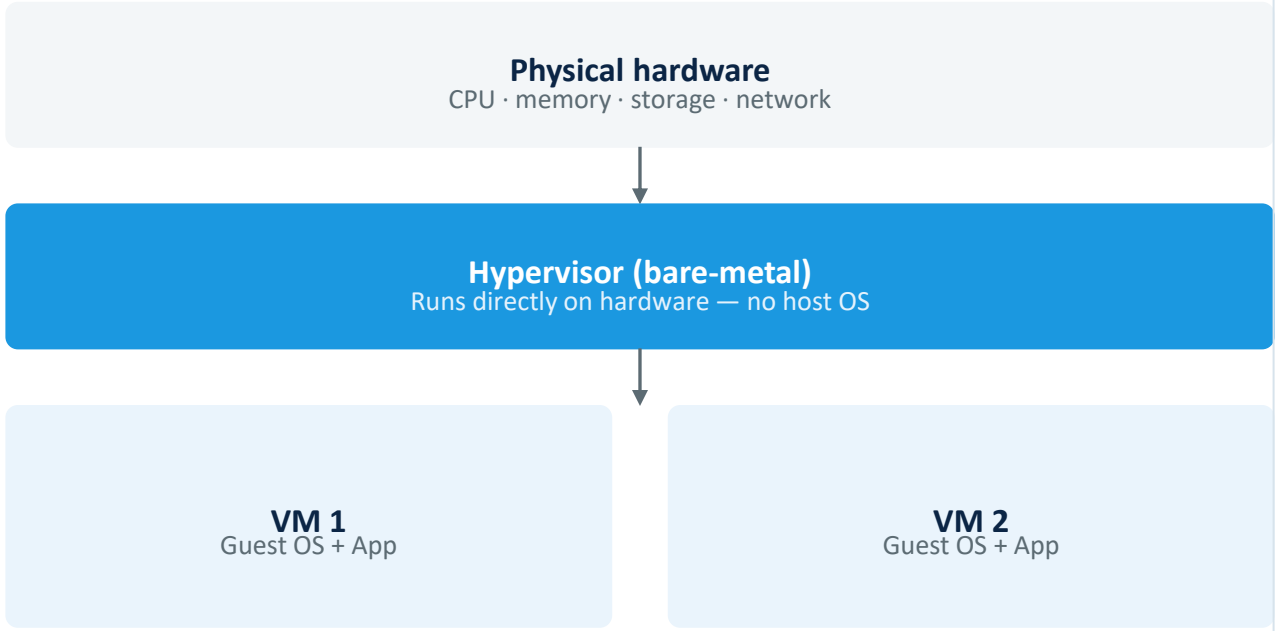
It allocates physical CPU, memory, storage, and network resources to each virtual machine and keeps them isolated from one another.

3

Two categories

Hypervisors are classified as Type 1 (bare-metal) or Type 2 (hosted), depending on what they run on.

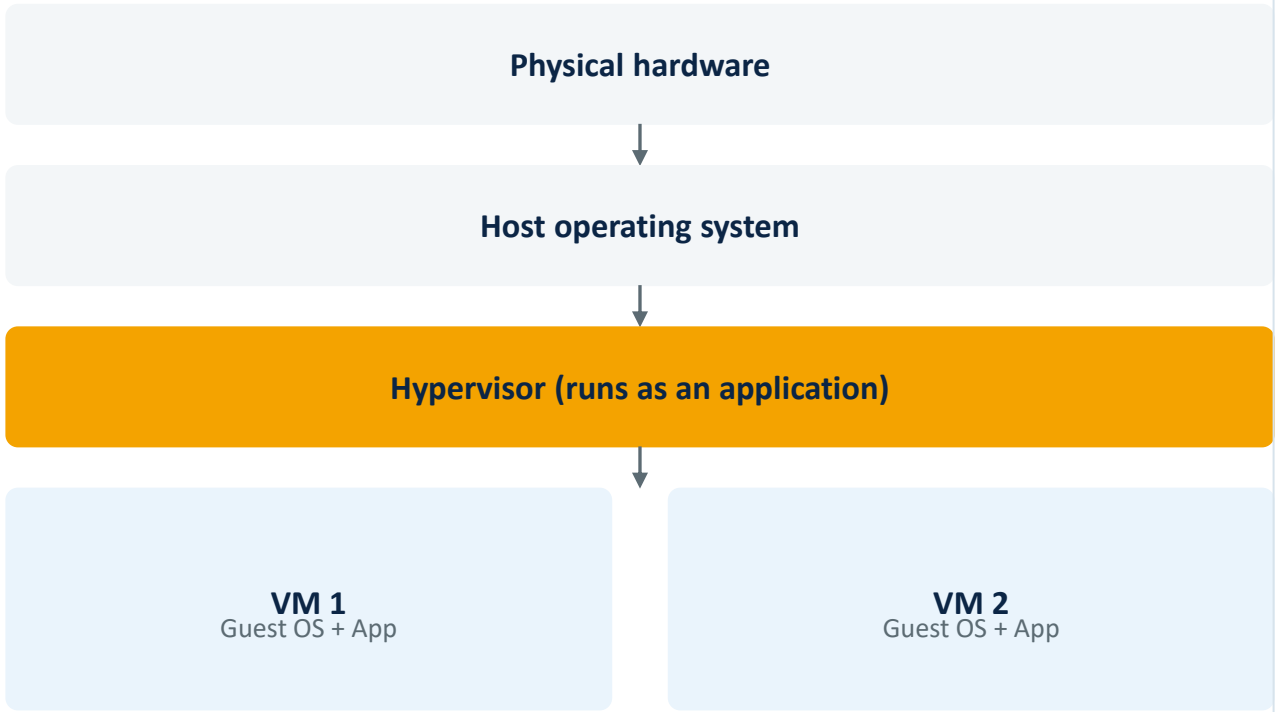
Type 1 Hypervisor (Bare-Metal)



Examples: VMware ESXi · Microsoft Hyper-V · Xen · KVM

- Runs directly on the physical hardware — there is no underlying host operating system.
- Communicates with hardware directly, so overhead is low and performance is close to native.
- Smaller attack surface than a hosted hypervisor, since there is no general-purpose host OS to secure.
- Standard choice for enterprise data centres and cloud platforms.

Type 2 Hypervisor (Hosted)



Examples: Oracle VirtualBox · VMware Workstation · Parallels Desktop

- Runs as an application on top of a conventional host operating system (Windows, macOS, Linux).
- Easier to install and use — it is just software installed like any other app.
- More overhead, since every hardware call passes through the host OS as well as the hypervisor.
- Common on developer laptops for testing, sandboxing, and personal use.

Type 1 vs Type 2: Comparison

Aspect	Type 1 (bare-metal)	Type 2 (hosted)
Runs on	Physical hardware directly	A host operating system
Performance	Near-native, low overhead	Lower, extra OS layer
Installation	More involved — replaces the OS	Simple — installed like an app
Typical use	Data centres, cloud platforms	Development, testing, personal use
Security	Smaller attack surface	Inherits host OS vulnerabilities
Examples	VMware ESXi, Hyper-V, Xen, KVM	VirtualBox, VMware Workstation, Parallels

How Hypervisors Virtualize Resources

1

CPU virtualization

Physical CPU cores are time-sliced and scheduled across multiple virtual CPUs, one or more per VM.

2

Memory virtualization

Physical RAM is partitioned and mapped so each VM has its own isolated, addressable memory space.

3

I/O virtualization

Virtual NICs and virtual disks stand in for physical network cards and storage devices, translating requests to the real hardware.

Types of Virtualization

1

Server virtualization

Multiple virtual servers run on one physical machine, each with its own OS.

2

Storage virtualization

Physical storage devices are pooled into a single logical storage resource.

3

Network virtualization

Network resources are abstracted into virtual switches, VLANs, and software-defined networks.

4

Desktop virtualization

Desktop operating systems run centrally and are delivered remotely to end-user devices (VDI).

Server Virtualization

1

Multiple VMs on one physical server

Each virtual machine runs its own guest operating system and applications, unaware of the others.

2

Consolidation

Many underused physical servers can be consolidated onto fewer, more heavily utilized machines.

3

Isolation

A failure or crash inside one VM does not directly affect the others running on the same host.

Storage Virtualization

1

Pooling physical storage

Disks and storage arrays from multiple devices are combined into one logical storage pool.

2

Simplified management

Administrators manage capacity as a single resource rather than tracking individual physical disks.

3

Advanced features

Enables snapshots, replication, thin provisioning, and easier backup and disaster recovery.

Network Virtualization

1

Abstracting the network

Virtual switches, routers, and VLANs are created independently of the underlying physical network hardware.

2

Software-defined networking (SDN)

Network control is centralized in software, separate from the physical devices that forward traffic.

3

Flexible traffic management

Virtual networks can be created, reconfigured, or torn down in software, without rewiring physical cables.

Benefits of Virtualization

1

Better hardware utilization

Multiple workloads share the same physical resources instead of sitting mostly idle.

2

Cost savings

Fewer physical servers means lower spending on hardware, power, and cooling.

3

Faster provisioning

New virtual machines can be created in minutes instead of ordering and racking new hardware.

4

Easier disaster recovery

VM snapshots and images can be backed up, copied, and restored quickly.

Limitations of Virtualization

1

Performance overhead

The hypervisor layer introduces some CPU, memory, and I/O overhead compared to running on bare hardware.

2

Licensing complexity

Software licensing models were not always designed with virtualized, movable workloads in mind.

3

Single point of failure

If the physical host or hypervisor fails, every VM running on it is affected simultaneously.

4

Security risks

A compromised hypervisor can potentially expose every VM it hosts — an attack sometimes called "hyperjacking".

2

PART 2

Virtual Machines vs Containers

Two different ways to package and isolate applications on shared infrastructure.

What is a Virtual Machine?

1

A complete virtual computer

Includes a full guest operating system, virtual CPU, memory, storage, and network interfaces.

2

Managed by a hypervisor

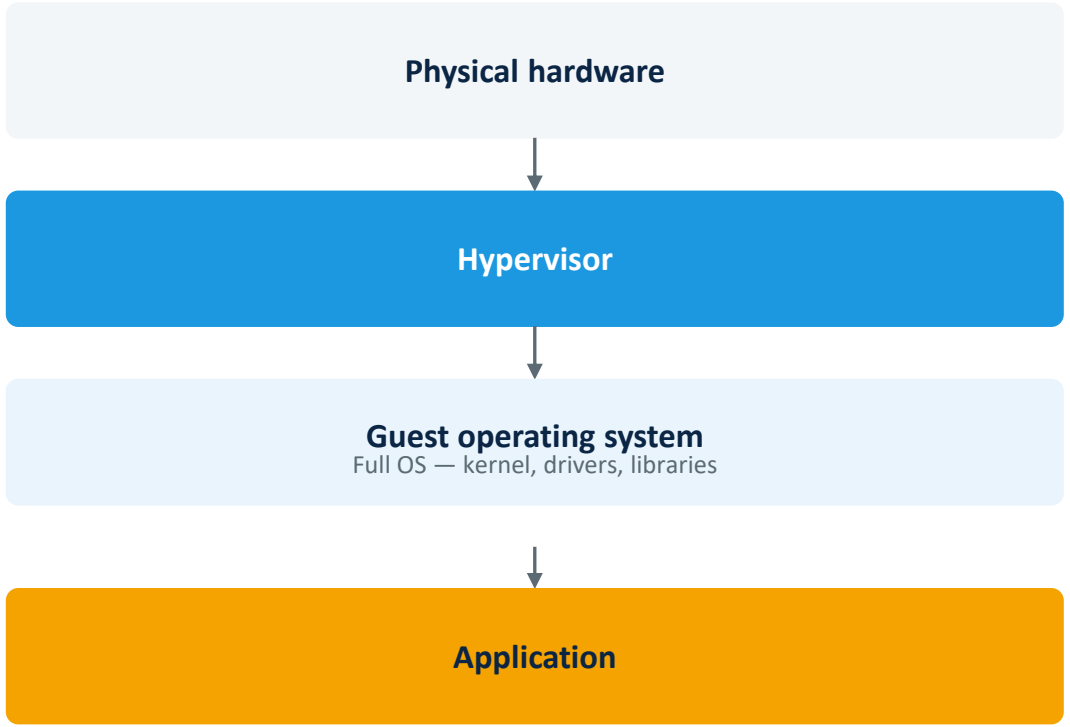
The hypervisor allocates physical resources to the VM and keeps it isolated from other VMs on the same host.

3

Strong isolation, heavier footprint

Each VM is independent and secure, but carries the size and boot time of a full operating system.

VM Architecture



A virtual machine packages a full guest OS with the app — strong isolation, heavier footprint, minutes to boot.

What is a Container?

1

A lightweight, portable package

Bundles an application together with just the code, runtime, and libraries it needs to run.

2

Shares the host OS kernel

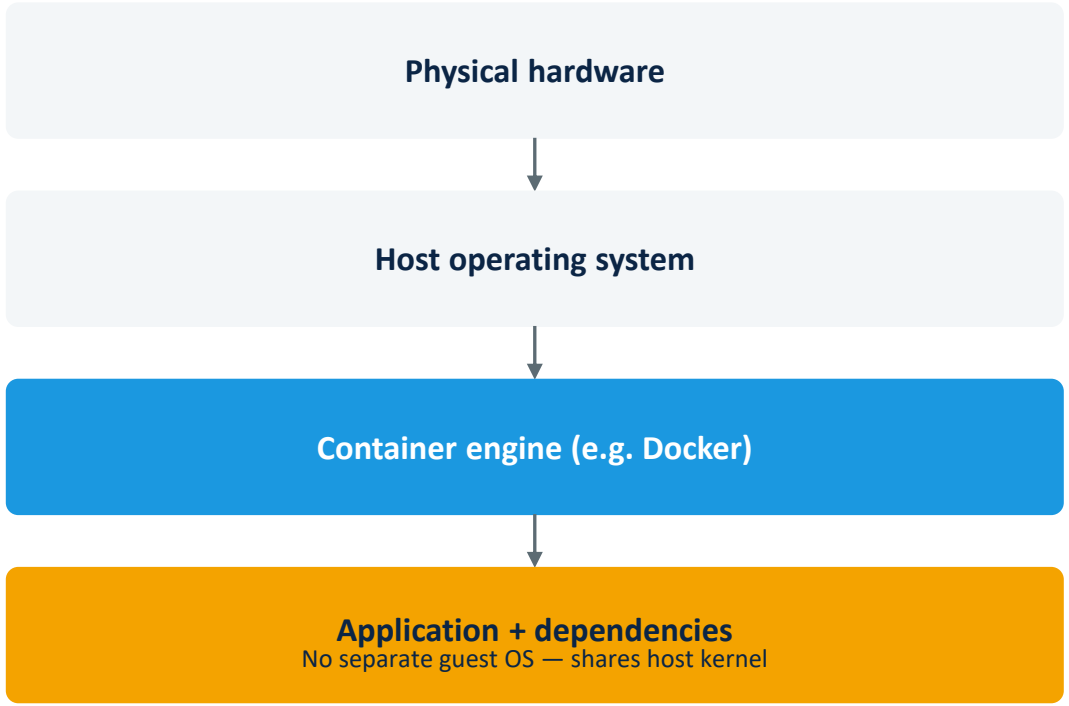
Unlike a VM, a container does not include its own operating system — it uses the kernel of the host it runs on.

3

Managed by a container engine

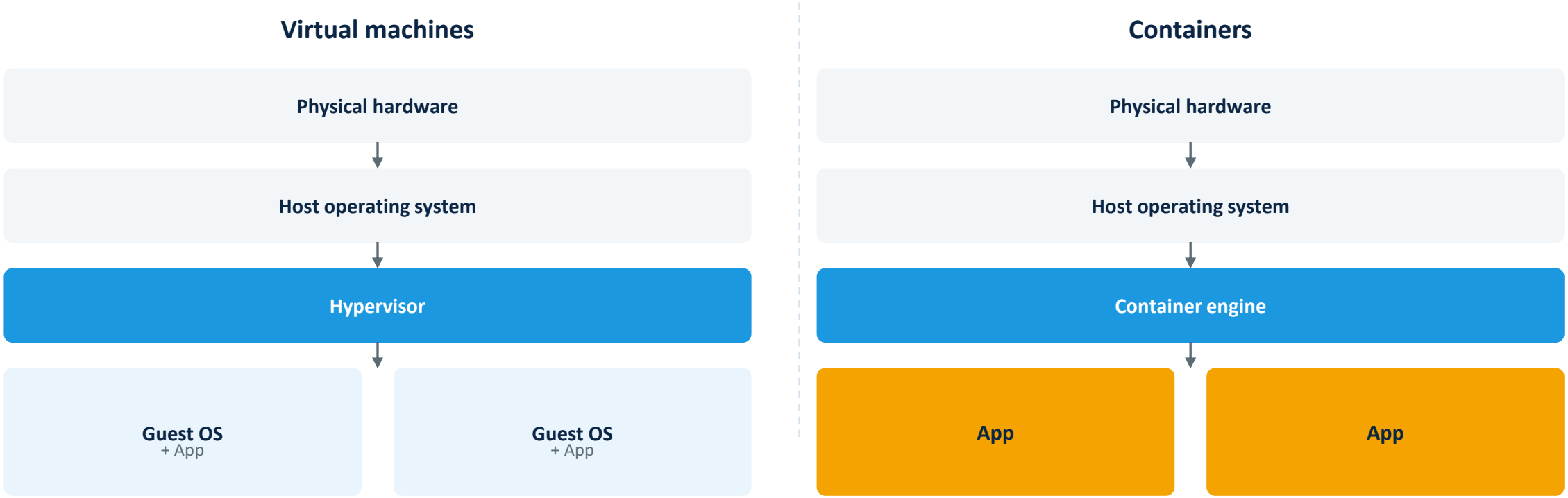
Engines such as Docker create, run, and manage containers on top of the host operating system.

Container Architecture



A container packages only the app and its dependencies, sharing the host kernel — lightweight, fast, seconds to start.

Architecture Compared



VM vs Container: Key Metrics

Metric	Virtual machines	Containers
Isolation	Strong — separate kernel	Weaker — shared kernel
Size	Gigabytes	Megabytes
Boot time	Minutes	Seconds
Density per host	Lower	Higher
Typical use	Strict isolation, mixed OS needs	Microservices, fast scaling, CI/CD

Docker & Container Engines

1

Docker

The most widely used container platform, providing tools to build, ship, and run containers.

2

Images and containers

A Docker image is a read-only template; a container is a running instance created from that image.

3

Dockerfile and registries

A Dockerfile scripts how an image is built; registries such as Docker Hub store and distribute images.

Choosing VMs vs Containers

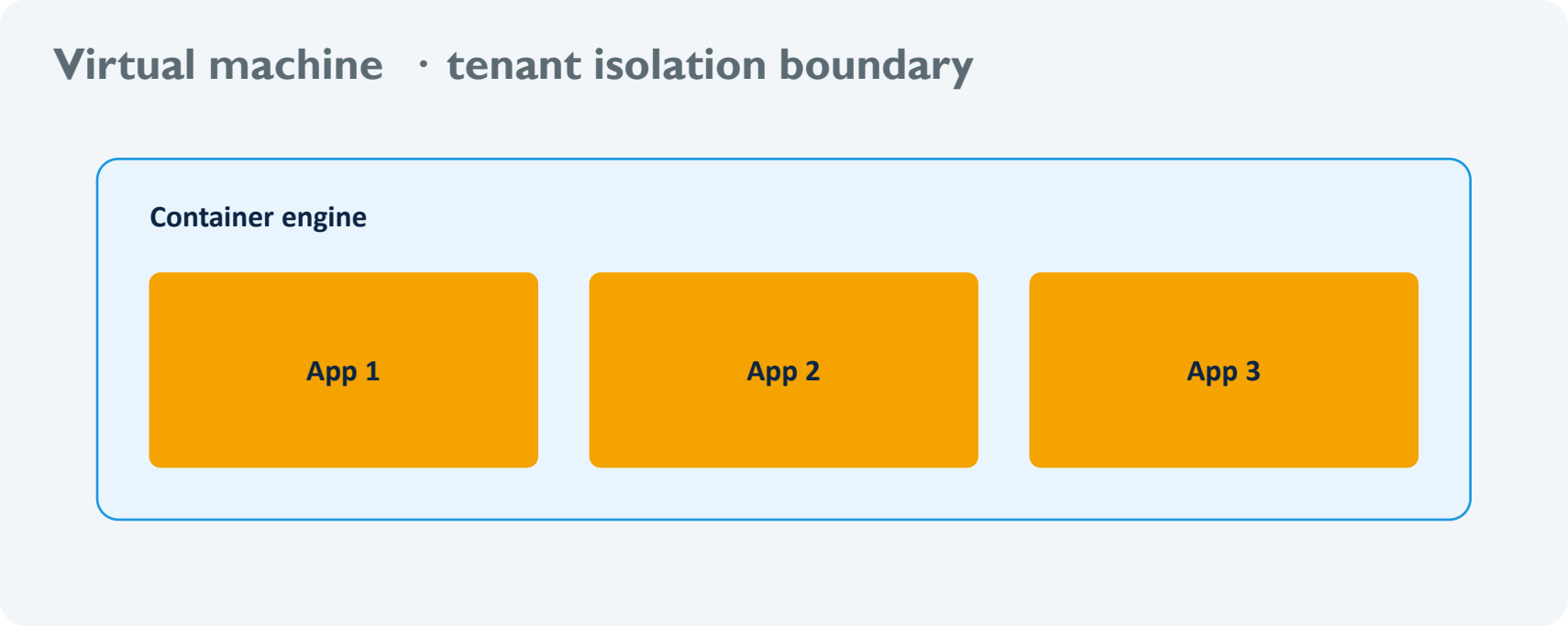
Use virtual machines when...

- Strong, hardware-level isolation between workloads is required.
- Different guest operating systems must run on the same host.
- Legacy applications depend on a specific full OS environment.
- Multi-tenant security boundaries are the top priority.

Use containers when...

- The workload is built as microservices.
- Fast startup and high deployment density matter.
- Applications are packaged and shipped through CI/CD pipelines.
- Elastic, rapid horizontal scaling is required.

Hybrid: Containers Inside VMs



Cloud providers commonly run containers inside a virtual machine: the VM supplies strong, hardware-enforced isolation between tenants, while containers inside it give fast, efficient deployment.

PART 3

3

Cloud Data Centre Architecture

How providers pool physical infrastructure and allocate it on demand.

What is a Data Centre?

1

Definition

A physical facility that houses the compute, storage, and networking infrastructure that powers cloud services.

2

Owned by providers or enterprises

Major cloud providers such as AWS, Microsoft Azure, and Google Cloud operate their own global networks of data centres.

3

The physical foundation of the cloud

Every virtual resource a customer uses ultimately runs on hardware inside one of these facilities.

Data Centre Physical Components

1

Servers and racks

Compute hardware organized into racks and clusters.

2

Storage arrays

Disk and flash storage systems holding customer and system data.

3

Networking gear

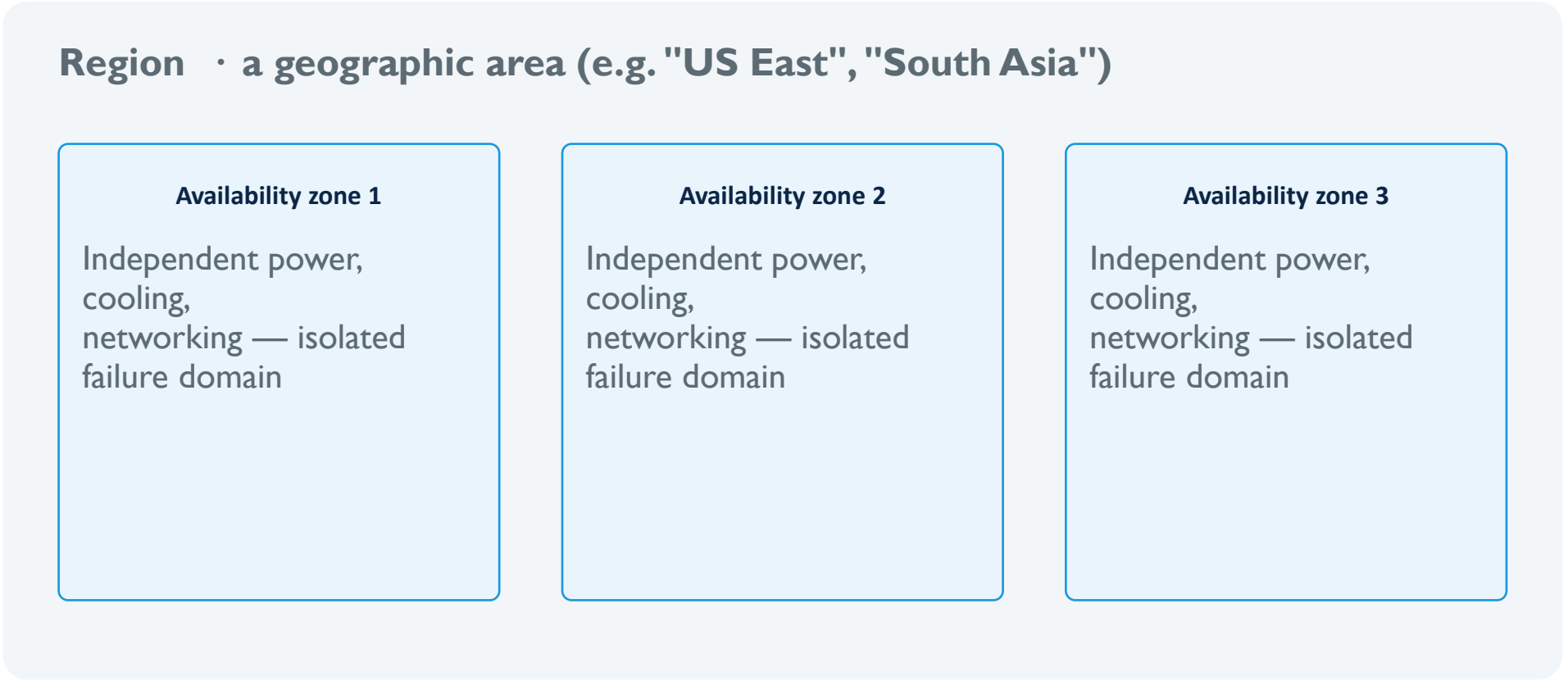
Switches, routers, and cabling connecting servers to each other and the internet.

4

Power and cooling

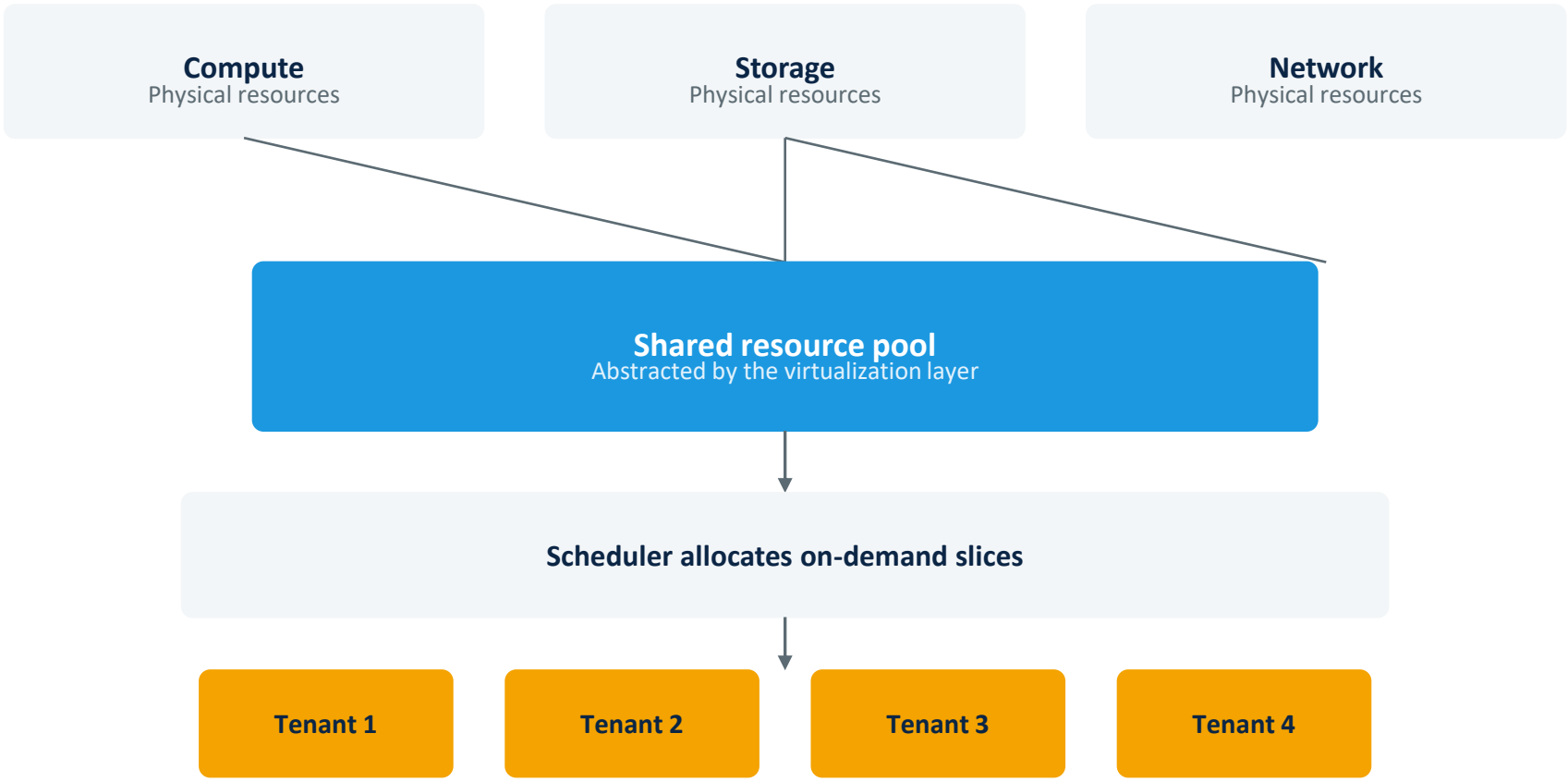
UPS systems, generators, and cooling infrastructure keeping hardware running reliably.

Regions & Availability Zones



Workloads are spread across zones so a single facility failure does not take the whole application offline.

Resource Pooling



Multi-Tenancy

1

Shared infrastructure, separate customers

Multiple customers ("tenants") run workloads on the same underlying physical infrastructure at the same time.

2

Isolation is enforced in software

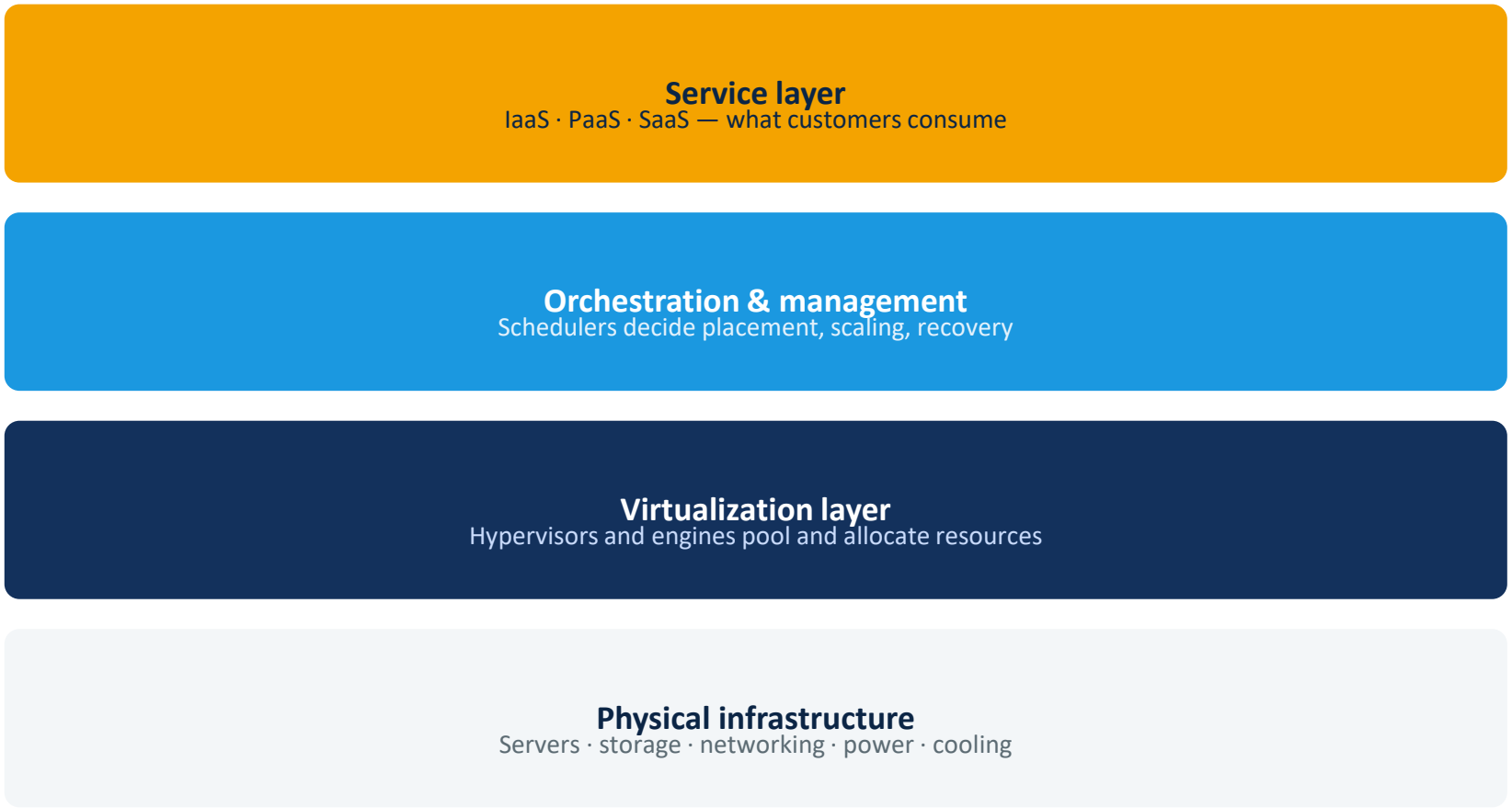
The virtualization layer keeps each tenant's compute, storage, and data securely separated from every other tenant.

3

Economies of scale

Sharing infrastructure across many tenants is what allows cloud providers to offer resources at low, pay-as-you-go cost.

Cloud Architecture Layers



Orchestration: Kubernetes

1

What it is

An open-source platform for automating the deployment, scaling, and management of containerized applications.

2

Scheduling workloads

Kubernetes decides which physical or virtual machine each container should run on, based on available resources.

3

Self-healing and scaling

It automatically restarts failed containers, scales replicas up or down, and redistributes workloads after a failure.

Elasticity & On-Demand Scaling

1

Dynamic reassignment

Pooled resources are reassigned to workloads automatically as demand rises and falls.

2

Auto-scaling

New VM or container instances are added automatically under load, and removed again once demand drops.

3

Pay-as-you-go economics

Customers pay only for the resources they actually consume, rather than for fixed, permanently owned capacity.

Real-World Example: Major Cloud Providers

Provider	Underlying virtualization	Global footprint
Amazon Web Services	Custom KVM-based Nitro hypervisor	Regions and Availability Zones worldwide
Microsoft Azure	Microsoft Hyper-V	Regions with paired-region redundancy
Google Cloud Platform	KVM-based hypervisor	Regions and zones across multiple continents

WRAP-UP

Summary / Key Takeaways

1

Virtualization concepts

Hypervisors (Type 1 bare-metal vs Type 2 hosted) abstract hardware so multiple OSes can share one machine.

2

VMs vs containers

VMs virtualize the hardware with a full guest OS; containers virtualize the OS and share the host kernel.

3

Cloud data centre architecture

Providers pool physical resources across regions and availability zones, allocating them on demand to many tenants.

THANK YOU

Questions & Discussion

Cloud Infrastructure and Virtualization · Virtualization Concepts, VMs vs Containers, Data Centre Architecture

Cloud Infrastructure and Virtualization Quiz

